higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 4

(12041014)

8 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 If a person comes into contact with live exposed conductors or any electrical circuits they may be electrocuted.

Explain in your own words what you understand by the term *electrocution.* (2)

1.2 The severity of electric shock is determined by the amount of electric current flowing through the body.

With reference to electric shock, give the resistance value of the human skin:

1.2.1 When the skin is dry

1.2.2 When the skin is wet or moist

(2 × 1) (2)

1.3 Study the diagram in FIGURE 1 below and answer the questions.

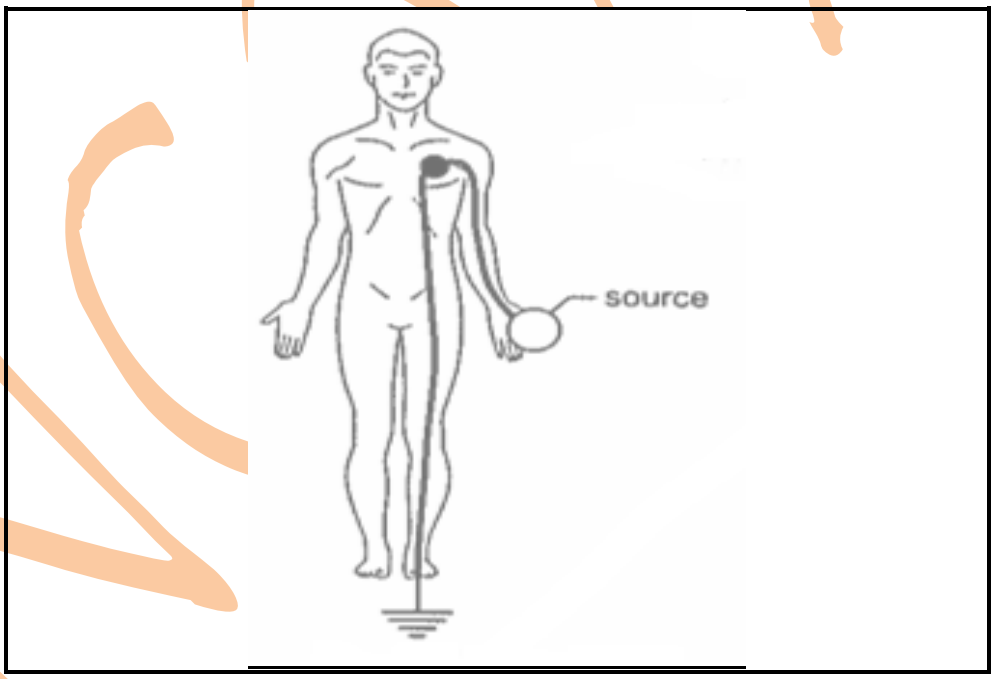


FIGURE 1

1.3.1 Identify the type of potential contact indicated. (2)

1.3.2 Explain the path that current travels through the body. (4)

1.4 Working with electrical equipment in the workplace can be very dangerous.

Name ONE safe work practice that should be used with regard to working with this equipment. (1 × 2) (2)

1.5 Name THREE main factors on which risk assessment in the workplace is based on. (3)

1.6 Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–D) next to the question number (1.6.1–1.6.3) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.6.1	Adjustable guards	A	are designed to be removed or opened to allow excess, e.g. to a hazard zone to insert or remove material from the point of operation
1.6.2	Self-adjusting guards	B	is a permanent part of the machine due to the need for it to be simple and effective
1.6.3	Interlocking guards	C	pushed away from the point of operation when material is fed into the machine; usually found on saws
		D	allow a machine to handle a wide variety of material sizes while still protecting the unused portion of the blade or the point of operation

(3 × 1) (3)

1.7 Explain the term *housekeeping*. (2)
[20]

QUESTION 2

2.1 Occupancy sensors are also referred to as motion sensors.

2.1.1 Name the TWO most common types of occupancy sensors available. (2)

2.1.2 Explain the operation of the TWO types of the occupancy sensors in question 2.1.1. (2 × 2) (4)

2.2 Draw a neat labelled circuit diagram of a three-heat stove switch to indicate how the switch is connected for medium heat operation. Make use of arrows on your diagram to indicate how conventional current flows. (5)

2.3 The operation of a geyser thermostat depends on the different rates of expansion of the invar rod and brass tube.

Briefly explain how the geyser thermostat operates to prevent current flow when the temperature of the water rises. (3)

2.4 Regulation 6.1 of SABS 0142 (SANS 1-10142) refers to generic circuit arrangements.

With regard to regulations 6.1.4 and 6.1.6, complete the sentences below by filling in the missing word(s). Write only the word(s) next to the question number (2.4.1–2.4.2) in the ANSWER BOOK.

2.4.1 All conductors of any circuit shall originate at the same ... … (2)

2.4.2 The … conductor shall not be connected directly to the earth or to the earth continuity conductor on the load side of the point of control. (1)

2.5 Protective devices must be provided to limit and break short-circuit currents before damage occurs on any electrical circuit.

Name THREE types of short-circuits associated with a three-phase system. (3)
[20]

QUESTION 3

3.1 What pocket-sized instrument is used by engineers to measure illuminance? (1)

3.2 *Quality of light* means that illumination from light should not be harmful to viewers.

Give TWO methods that could be used to avoid shadows when using lighting. (2)

3.3 Name ONE application or environment where each of the following light classifications is used:

3.3.1 Direct lighting

3.3.2 Indirect lighting

3.3.3 Semi-direct lighting

3.3.4 Semi-indirect lighting

(4 × 1) (4)

3.4 *Efficacy* is the lighting term used when describing how efficiently a lamp converts electrical energy into visible light.

3.4.1 Which type of lamp has the lowest luminous efficacy of all commonly used lamps? (1)

3.4.2 Name the TWO subgroups the lamp in QUESTION 3.4.1 above is divided into and which are available in different shapes and sizes. (2)

3.4.3 What is the average lifespan, in hours, of the lamp mentioned in QUESTION 3.4.1 above? (1)

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3.5 Light-emitting diodes are used as indicator lamps in many devices.

Name TWO applications of LED's. (2)

3.6 Explain what you understand by the term *stroboscopic effect*. (3)

3.7 The starter in a fluorescent lamp is wired in parallel to the tube and serves two functions.

3.7.1 Name TWO types of starters. (2)

3.7.2 State TWO functions of a starter. (2)

[20]

QUESTION 4

4.1 Class 1 and Class 11 appliances are two basic types of electrical appliances.

Explain the type of insulation for the following:

4.1.1 Class 1 appliances

4.1.2 Class 11 appliances

(2 × 2) (4)

4.2 Name THREE basic things one should know before performing any fault finding and repair work on domestic appliances or power tools. (3)

4.3 Name TWO factors that need to be considered when replacing cables on domestic appliances or power tools. (2)

4.4 The oven or grill on a stove is not working. You suspect the timer control to be faulty.

Explain the procedure you will follow to verify the operation of the timer control. (3)

4.5 Fault finding, repairs or maintenance of a customer's domestic appliances, power tools, equipment or any other electrical machinery can be done in various places.

Name THREE places where you would carry out any of the above work. (3)

4.6 Draw a neat labelled block diagram of a *fuzzy logic* control programme. (5)

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QUESTION 5

5.1 Planning plays an important part in the effective completion of any electrical task.

Explain in your own words, what is meant by the term *planning a task.* (2)

5.2 After installing an electrical motor, you will need to perform certain electrical tests on the motor.

Name FIVE basic electrical test instruments you will require to perform these tests. (5)

5.3 As a maintenance electrician you are required to perform a three-monthly schedule on a three-phase induction motor.

5.3.1 Explain how you would clean the motor. (6 × 1) (6)

5.3.2 State THREE electrical tests that you would perform on the stator windings of the motor. (3)

5.4 Proper sizing of electrical machines is important as the load determines how much power it draws.

Give FOUR reasons why machines are often oversized. (4)

[20]

TOTAL: 100

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